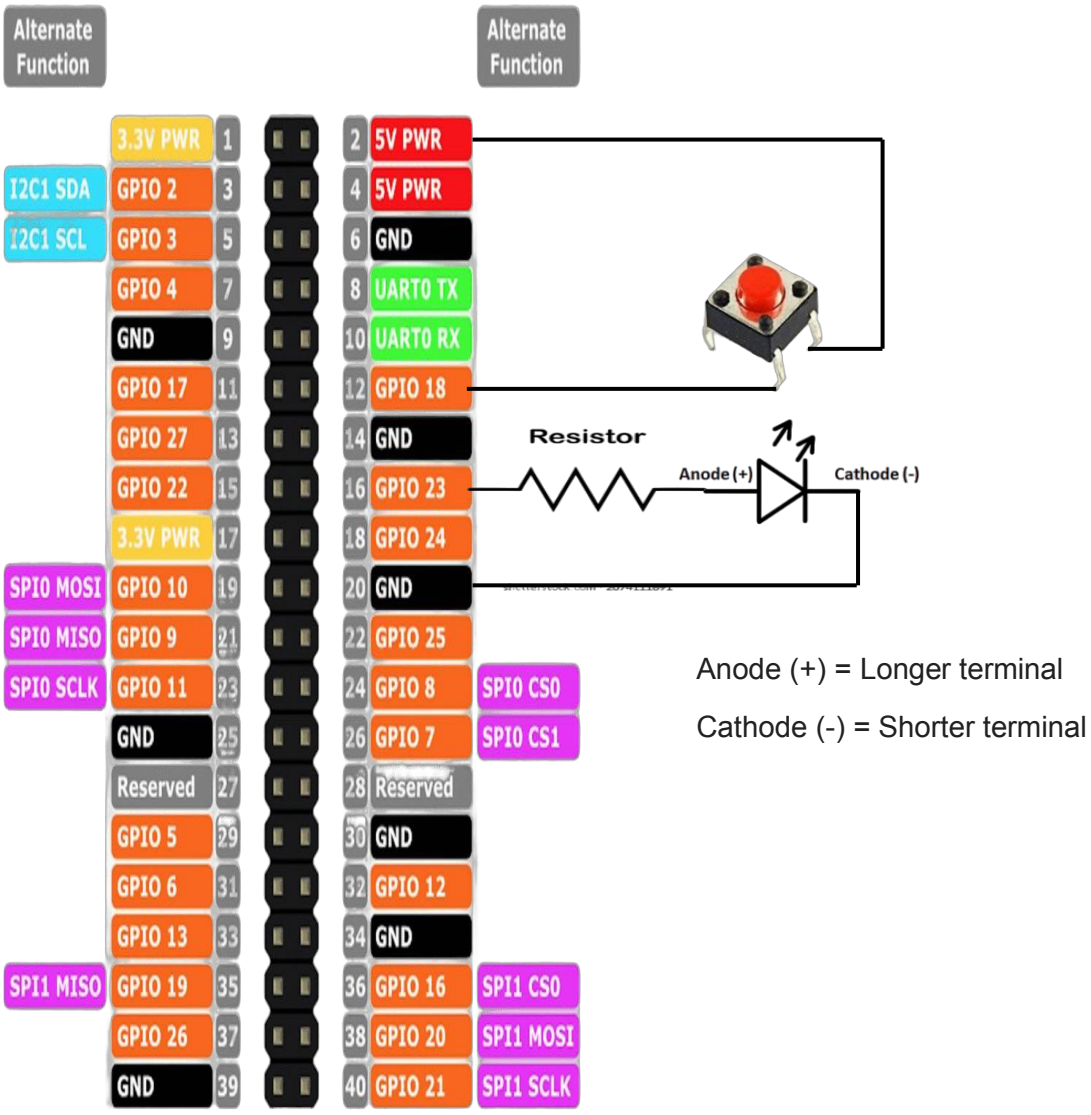
PRACTICAL 17 (Module 2)

TITLE : Design and implement a Node-RED flow on Raspberry Pi that integrates a physical push-button and LED with a web dashboard.

- **AIM :** To design and implement a Node-RED flow on Raspberry Pi that integrates a physical push-button and LED with a web dashboard. The flow must allow the users to toggle the LED from the dashboard and via the physical button.
- **HARDWARE COMPONENTS REQUIRED :** Raspberry Pi 400 (Pi version 4), power adaptor, HDMI to micro HDMI cable, mouse, monitor, push button, resistor, LED, jumper wires
- **CIRCUIT SCHEMATIC :**



Anode (+) = Longer terminal

Cathode (-) = Shorter terminal

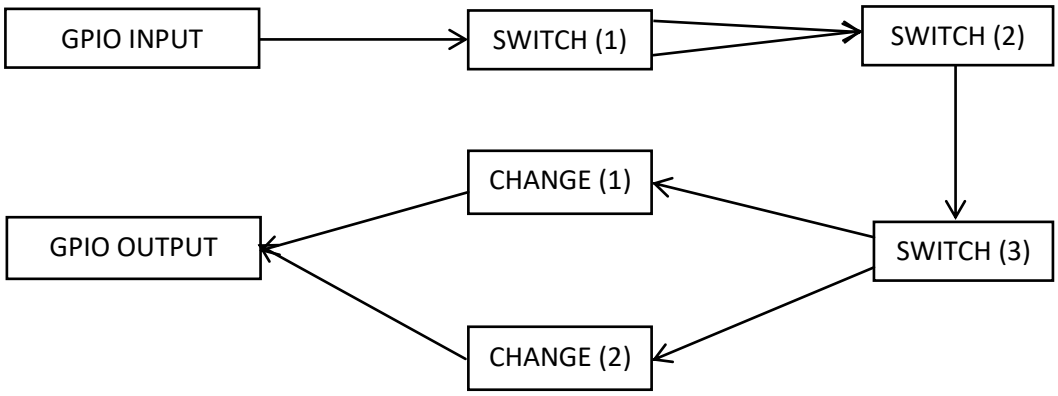
NOTE : Draw and show the connections of the push button, resistor and LED with Raspberry Pi as per the reference diagram, the GPIO pins could change as per your connections -

(Draw the exact same diagrams for the raspberry pi and the components as the image attached above with all the pin numbers and names)

- **COMMANDS :**

  - bash <(curl -sL https://github.com/node-red/linux-installers/releases/latest/download/install-update-nodered-deb)
    [To install Node RED on Raspberry Pi]
  - node-red-start [This starts the Node-RED service. Open the url displayed on the terminal]
  - node-red-stop [This stops the Node-RED service]
  - node-red-restart [This restarts the Node-RED service]
  - node-red-reload [This stops then starts the Node-RED service]

- **NODE RED FLOW** :



- **NODE PROPERTIES**

  - **GPIO input :** Select the GPIO pin number connected with the push button
  - **Switch (1) :** Property – global.state
    Rule 1 - ==0
    Rule 2 - ==1
  - **Switch (2) :** Property – msg.payload
    Rule - ==1
  - **Switch (3) :** Property – global.state
    Rule 1 - ==0
    Rule 2 - ==1

- **Change (1) :** Property – global.state (==1)
  msg.payload (==1)
- **Change (2) :** Property – global.state (==0)
  msg.payload (==0)
- **GPIO output :** Select the GPIO pin number connected with the LED

- **RESULS & CONCLUSION** : **Note –** Explain the output (If no output got - Mention the errors displayed)